

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
CONCEPT RECAPITULATION TEST – I
PAPER –2
TEST DATE: 30-06-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. ABD

2. BC

Sol. m, M will separate when a_x for 'm' becomes zero.

3. BC

Sol. For the bubble, $PV = \text{const.}$

$$\therefore \frac{dV}{V} = \frac{-dP}{P}$$

$$\frac{dV}{V} = \frac{3dr}{r}, dP \approx -\frac{4T}{R}, P \approx P_0 \text{ (according to the question)}$$

$$\therefore dr = \frac{4Tr}{3P_0R}$$

4. ACD

5. ABD

6. BC

7. A

8. D

Sol. **(for Q. 7 – 8)**

For the charge

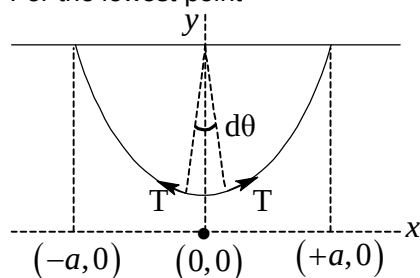
$$\vec{F} = q(\vec{v} \times \vec{B}) - b\vec{v}$$

$$\therefore \frac{d\vec{v}}{dt} = \frac{q}{m} \left(\frac{d\vec{r}}{dt} \times \vec{B} \right) - \frac{b}{m} \left(\frac{d\vec{r}}{dt} \right)$$

$$\therefore \Delta\vec{v} = \frac{q}{m} (\Delta\vec{r} \times \vec{B}) - \frac{b(\Delta\vec{r})}{m}$$

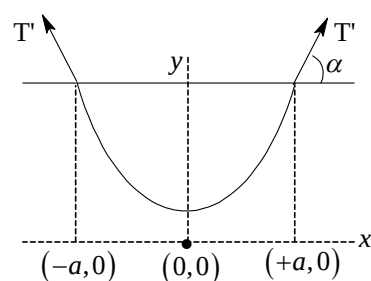
9. B

Sol. For the lowest point



$$2T \left(\frac{d\theta}{2} \right) = \mu R d\theta g$$

10. A

 Sol. $T' \cos \alpha = T$


Section – B

11. 6

12. 8

13. 1

Sol. In the CM frame, work done by pseudo forces $= 2 \left(\frac{F}{2} \times \ell \right) = F\ell$, this is the energy lost in collision.

Section – C

14. 0.60

15. 1.40

16. 17.90

17. 6.00

18. 100.00

19. 3.60

Chemistry**PART – II****Section – A**

20. AC

Sol. Ore that have magnetic properties are concentrated using magnetic separator. Chromite ore – FeCr_2O_4 , Ilmenite FeTiO_3 .

21. AB

Sol. (A) Mond's process proceeds through formation of $[\text{Ni}(\text{CO})_4]$
 (B) Froth flotation, Zn is deactivated in the presence of lead by addition of NaCN. The complex $\text{Na}_2[\text{Zn}(\text{CN})_4]$ is formed.

22. BC

Sol. Due to difference in number of ion formed, there will be a difference in the freezing point measurement. AgNO_3 test – one mole of each complex on reaction with excess AgNO_3 will give two moles and three moles of AgCl respectively.
 Since Co is in the same oxidation state and number of NH_3 ligands differ by one only there will be no difference in magnetic moment. MnO_2 test is not quantitative hence number of Cl^- ions cannot be identified.

23. BCD

Sol. If $\mu = 5.92 \text{ BM}$ with weak field ligand, the value of μ will be 1.73 BM in strong field ligand.

24. AC

Sol. (A) $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$ has 3 electrons hence t_{2g}^3 irrespective of ligand, pairing will not occur.
 (C) $[\text{TiCl}_6]^{2-}$ has a d^0 configuration so μ is always zero.

25. ACD

Sol. $\text{Na}_2\text{O}_2 + \text{CO}_2 \longrightarrow \text{Na}_2\text{CO}_3 + \frac{1}{2}\text{O}_2$

26. B

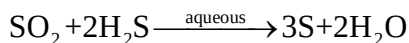
27. B

28. B

29. A

Sol. $2\text{SO}_2 \rightleftharpoons \text{SO}^{2+} + \text{SO}_3^{2-}$

A is SO_2  sp^2 hybridized sulphur B is H_2S  sp^3 hybridized



A B

Section – B

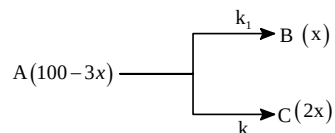
30. 268

31. 6

Sol. From the given reaction, we have $t_{1/2}$ of B = $2t_{1/2}$ of C, which means that C is formed at twice the rate of formation of B. The rate constant for the decay of A is.

$$k=k_1+k_2=\frac{0.693}{9}+\frac{0.693}{4.5}=\frac{0.693}{3}$$

Let after time t , $[C] = 2[A]$ and $[B] = x\%$. Thus, the reaction can be represented as



Now, for first order kinetics, we have

$$kt = \ln \frac{a}{a-x} \Rightarrow \frac{\ln 2}{3} t = \ln \frac{100}{100-3x} \Rightarrow 2^{t/3} = \frac{100}{25} = 4$$

Therefore, $\frac{l}{3} = 2 \Rightarrow t = 6$

32. 6

Sol. $\text{KMnO}_4 + \text{KI} \longrightarrow \text{MnO}_2 + \text{I}_2$

$$\text{Eq. of KMnO}_4 = \text{Eq. of I}_2$$

$$4 \times 3 = n \times 2$$

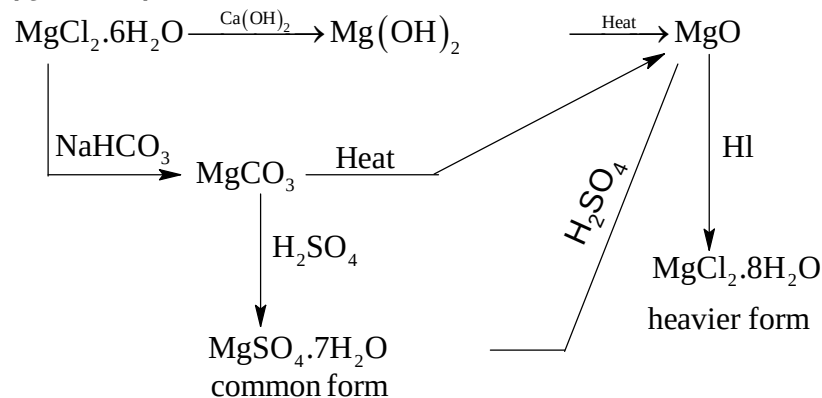
n=6

Section – C

33. 422.00

34. 9.76

Sol. (Q. 33 – 34)



35. 50.00

36. 99.83

Sol. For the precipitation, the ionic product should be greater than solubility product. Also, K_{sp} of $Hg_2I_2 = 2.5 \times 10^{-26}$ is much less than of $AgI = 8.5 \times 10^{-17}$.

For the reaction $Hg_2I_2 \rightleftharpoons Hg_2^{2+} + 2I^-$, the solubility product is

$$K_{sp} = [Hg_2^{2+}][I^-]^2 = 2.5 \times 10^{-26}$$

$$\text{Therefore, } [I^-] = \sqrt{\frac{K_{sp}}{[Hg_2^{2+}]}} = \sqrt{\frac{2.5 \times 10^{-26}}{0.1}} = 5.0 \times 10^{-13} M$$

For the reaction $AgI \rightleftharpoons Ag^+ + I^-$, the solubility product is

$$K_{sp} = [Ag^+][I^-] = 8.5 \times 10^{-17}$$

$$\text{Therefore, } [I^-] = \frac{K_{sp}}{[Ag^+]} = \frac{8.5 \times 10^{-17}}{0.1} = 8.5 \times 10^{-16} M$$

Therefore,

$$[Ag^+] = \frac{8.5 \times 10^{-17}}{5.0 \times 10^{-13}} = 1.7 \times 10^{-4} M$$

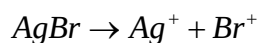
So, the % of Ag^+ precipitated is $100 - 0.7 = 99.83\%$

37. 10.00

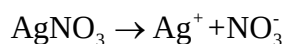
38. 62.00

Sol. Suppose solubility of $AgBr$ in presence of $10^{-7} M AgNO_3$ solution is S .

Then the reactions involved are



$S \qquad S$



The concentration terms $[Ag^+] = [S + 10^{-3}]$ and $[Br^-] = S$. Now, $K_{sp} = [Ag^+][Br^-]$,

substituting, we get $12 \times 10^{-14} = (S + 10^{-7})S \Rightarrow S^2 + 10^{-7}S + 12 \times 10^{-14} = 0$ which is equation in S , solving and ignoring negative root, we get

$$S = \frac{-10^{-7} + \sqrt{10^{-14} + 4 \times 12 \times 10^{-14}}}{2}$$

$$= \frac{-10^{-7} + 7 \times 10^{-7}}{2} = 3 \times 10^{-7} M$$

In solution, we have

$$[Ag^+] = S + 10^{-7} = 3 \times 10^{-7} + 10^{-7} = 4 \times 10^{-7} M; [Br^-] = S = 3 \times 10^{-7} M; [NO_3^-] = 10^{-7} M,$$

Now multiply by 1000 to convert M to mol/m^3

Now specific conductance is

$$\kappa = \kappa_{Ag^+} + \kappa_{NO_3^-} = \lambda_{(Ag^+)}^0 \times C_{(Ag^+)} + \lambda_{(Br^-)}^0 \times C_{(Br^-)} + \lambda_{(NO_3^-)}^0 \times C_{(NO_3^-)}$$

$$= 6 \times 10^{-3} \times 4 \times 10^{-7} + 8 \times 10^{-3} \times 3 \times 10^{-7} + 7 \times 10^{-3} \times 10^{-7}$$

$$= 55 \times 10^{-10} S m^{-1}$$

Mathematics

PART – III

Section – A

39. ABC

Sol. $3x^2 \frac{dx}{dy} = \frac{x^3}{x^3 - y} \quad x^3 = t$

$$\Rightarrow \frac{dt}{dy} = \frac{t}{t - y} \Rightarrow (t - y)dt = t dy$$

$$\Rightarrow t dt = y dt + t dy \Rightarrow \frac{t^2}{2} = t.y + \frac{c}{2}$$

$$\Rightarrow x^6 = 2x^3 y + c$$

40. ABC

Sol. The 7-digit Numbers with 1 in the left most place and containing each of the digits 1, 2, 3, 4, 5, 6, 7 exactly once is $6! = 720$.

41. ABCD

Sol. $f(x) = \begin{cases} px + \lambda, & x > 0 \\ q, & x = 0 \\ e^x, & x < 0 \end{cases}$

42. AC

Sol. $f(x) = (2x - 1)(2x + 1)(x - 2)(x - 4)$

43. BCD

Sol. $p(n)$ = Probability that child stops after drawing exactly n marbles

$$p(n) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdots \left(\frac{n-1}{n} \right) \left(\frac{1}{n+1} \right) = \frac{1}{n(n+1)}$$

44. CD

Sol. If α is real root then $\alpha = 2, m = \pm 2$, non-real root = $1 + i$

45. A

Sol. For $\lambda = 10 \Rightarrow t_1 = 2, t_2 > 2$ and if $x + 1/x = 2 \Rightarrow x = 1$ and $x + 1/x = t_2$ so there will be two distinct values of x

46. B

Sol. Since domain of $f(t) = 2t^3 - 9t^2 + 30, t = x + 1/x, |t| \geq 2, f(-2) = -22, f(2) = 10$, critical points at $t = 0 \& 3; f(3) = 3$

47. A

48. D

Section – B

49. 6

Sol. $p(x) = (x^2 - 4)(x^4 - 10x^2 + 10)$

50. 6

Sol. $6 + 4 - 4$

51. 6

Sol. $\lim_{x \rightarrow 0} \frac{\ln \cos 2x}{x^2} = -\frac{2^2}{2}, \lim_{x \rightarrow 0} \frac{\ln \cos 4x}{x^2} = -\frac{4^2}{2}$

Section – C

52. 0.33

53. 0.01

54. 4.00

55. 3.00

56. 6.00

57. 3.00